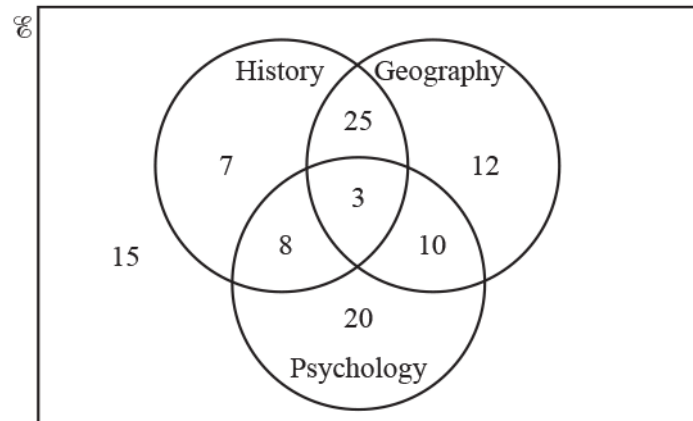


- 1 The Venn diagram shows the numbers of students studying various subjects, in a year group of 100 students.



A student is chosen at random from the 100 students. Then another student is chosen from the remaining students.

Find the probability that the first student studies History and the second student studies Geography but not Psychology.

[4]

2(a) Anika and Beth are playing a game which consists of several points.

- The probability that Anika will win any point is 0.7.
- The probability that Beth will win any point is 0.3.
- The outcome of each point is independent of the outcome of every other point.

The first player to win two points wins the game.

Write down the probability that the game consists of more than three points.

[1]

(b) Draw a tree diagram to show all the possibilities for the game.

[3]

(c) Determine the probability that Beth wins the game.

[3]

(d) Determine the probability that the game consists of exactly three points.

[2]

(e) Given that Beth wins the game, determine the probability that the game consists of exactly three points.

[4]

3(a) Jane conducted a survey. She chose a sample of people from three towns, A, B and C. She noted the following information.

400 people were chosen.

230 people were adults.

55 adults were from town A.

65 children were from town A.

35 children were from town B.

150 people were from town B.

Complete the two-way frequency table. [2]

		Town			Total
		A	B	C	
adult					
child					
Total					

(b) One of the people is chosen at random.

(i) Find the probability that this person is an adult from town A.

[1]

(ii) Given that the person is from town A, find the probability that the person is an adult.

[1]

- (c) For another survey, Jane wanted to choose a random sample from the 820 students living in a particular hostel. She numbered the students from 1 to 820 and then generated some random numbers on her calculator.

The random numbers were 0.114 287 562 and 0.081 859 817.

Jane's friend Kareem used these figures to write down the following sample of five student numbers.

114, 142, 428, 287 and 756

Jane used the same figures to write down the following sample of five student numbers.

114, 287, 562, 81 and 817

- (i) State, with a reason, which one of these samples is not random.

[1]

- (ii) Explain why Jane omitted the number 859 from her sample.

[1]

4(a) In this question you must show detailed reasoning.

The random variable X has probability distribution defined as follows.

$$P(X = x) = \begin{cases} \frac{15}{64} \times \frac{2^x}{x!} & x = 2, 3, 4, 5, \\ 0 & \text{otherwise.} \end{cases}$$

Show that $P(X = 2) = \frac{15}{32}$.

[1]

(b) The values of three independent observations of X are denoted by X_1 , X_2 and X_3 .

Given that $X_1 + X_2 + X_3 = 9$, determine the probability that at least one of these three values is equal to 2. [6]

(c) Freda chooses values of X at random until she has obtained $X = 2$ exactly three times. She then stops.

Determine the probability that she chooses exactly 10 values of X . [3]

5(a) A random variable X has probability distribution defined as follows.

$$P(X = x) = \begin{cases} kx & x = 1, 2, 3, 4, 5, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

Show that $P(X = 3) = 0.2$.

[3]

(b) Show in a table the values of X and their probabilities.

[2]

(c) Two independent values of X are chosen, and their total T is found.

(i) Find $P(T = 7)$.

[3]

(ii) Given that $T = 7$, determine the probability that one of the values of X is 2.

[4]

- 6 A counter is initially at point O on the x -axis. A fair coin is thrown 6 times. Each time the coin shows heads, the counter is moved one unit in the positive x -direction. Each time the coin shows tails, the counter is moved one unit in the negative x -direction. The final distance of the counter from O , in either direction, is denoted by D .

Determine the most probable value of D .

[7]

- 7 Rob has two six-sided dice, each with sides numbered 1, 2, 3, 4, 5, 6.
One dice is fair. The other dice is biased, with probabilities as shown in the table.

Biased die						
y	1	2	3	4	5	6
$P(Y = y)$	0.3	0.25	0.2	0.14	0.1	0.01

Rob throws each dice once and notes the two scores, X on the fair dice and Y on the biased dice. He then calculates the value of the variable S which is defined as follows.

- If $X \leq 3$, then $S = X + 2Y$.
- If $X > 3$, then $S = X + Y$.

- (a) (i) Draw up a sample space diagram showing all the possible outcomes and the corresponding values of S . [2]

- (ii) On your diagram, circle the four cells where the value $S = 10$ occurs.

[1]

(b) Explain the mistake in the following calculation.

[1]

$$P(S = 10) = \frac{\text{Number of outcomes giving } S = 10}{\text{Total number of outcomes}} = \frac{4}{36} = \frac{1}{9}.$$

(c) Find the correct value of $P(S = 10)$.

[2]

(d) Given that $S = 10$, find the probability that the score on one of the dice is 4.

[3]

(e) The events “ $X = 1$ or 2 ” and “ $S = n$ ” are mutually exclusive. Given that $P(S = n) \neq 0$, find the value of n . [1]

8

- (a) Events A and B are independent, and $P(A \cap B) = \frac{1}{24}$ and $P(A \cup B) = \frac{3}{8}$.

Find $P(A)$ and $P(B)$.

[5]

- (b) Events C and D are such that $P(C) = 0.6$, $P(D) = 0.3$ and $P(C \cup D) = 0.8$. Find $P(D | C)$.

[4]

9 Each of the 30 students in a class plays at least one of squash, hockey and tennis.

- 18 students play squash
- 19 students play hockey
- 17 students play tennis
- 8 students play squash and hockey
- 9 students play hockey and tennis
- 11 students play squash and tennis

(a) Find the number of students who play all three sports.

[3]

A student is picked at random from the class.

(b) Given that this student plays squash, find the probability that this student does not play hockey.

[1]

Two different students are picked at random from the class, one after the other, without replacement.

(c) Given that the first student plays squash, find the probability that the second student plays hockey.

[4]

10 A random variable X has probability distribution given by

$$P(X = x) = \frac{1}{860}(1 + x) \quad \text{for } x = 1, 2, 3, \dots, 40.$$

(a) Find $P(X > 39)$.

[2]

(b) Given that x is even, determine $P(X < 10)$.

[6]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			P(1st HGP' & 2nd GP' or 1st HGP & 2nd GP' or 1st HG' & 2nd GP') $= \frac{25}{100} \times \frac{36}{99} + \frac{3}{100} \times \frac{37}{99} + \frac{15}{100} \times \frac{37}{99}$ $= \frac{87}{550} \text{ or } 0.158 \text{ (3 sf)}$	M1(AO3.1b) M1(AO1.1) M1(AO2.1) A1(AO1.1) [4]	One of the following expressions seen: M1 Another seen: M1 $\frac{25}{100} \times \frac{\dots}{99}$ or $\frac{3}{100} \times \frac{\dots}{99}$ or $\frac{15}{100} \times \frac{\dots}{99}$ or $(\frac{8}{100} \times \frac{\dots}{99} + \frac{7}{100} \times \frac{\dots}{99})$ 3rd M1 for all correct Examiner's Comments Most candidates found P(1st studies History) $\times$ P(2nd studies Geography but not Psychology) $= \frac{43}{100} \times \frac{37}{99}$ or possibly $\frac{43}{100} \times \frac{36}{99}$. These methods do not take into account the possible overlaps between the two events. The two events need to be separated into different cases. In order to gain any marks, candidates needed to show products such as $\frac{25}{100} \times \frac{36}{99}$ or $\frac{3}{100} \times \frac{37}{99}$. Full details can be found in the published mark scheme.
			Total	4	
2	a		0	B1 [1]	Allow 0%

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b			B1 B1 B1 [3]	Ignore extra branches if no probabilities or $p = 0$ B2: 8 correct branches and probs <u>OR</u> names, no extra branches B2: 7 correct branches, probs and names, no extra branches B1: 8 correct branches without probs & names. No extra branches B1: 6 correct branches, probs and names. Ignore extra branches Ignore products at ends
	c		$0.3 \times 0.3 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.3$ or $0.09 + 0.063 + 0.063$ oe $= \frac{27}{125}$ or 0.216	M1 M1 A1 [3]	All correct M2 ft their diagram Two products correct M1 ft their diagram SC Correct answer with no working: B2
	d		$0.7 \times 0.3 + 0.3 \times 0.7$ or $0.21 + 0.21$ $= \frac{21}{50}$ or 0.42	M1 A1 [2]	or $1 - (0.7^2 + 0.3^2)$ or $1 - (0.49 + 0.09)$ Condone missing brackets or $0.7 \times 0.3 \times 0.7 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.7 + 0.3 \times 0.7 \times 0.3$ oe or $2 \times 0.147 + 2 \times 0.063$ oe Wholly correct method ft their diagram SC Correct answer with no working: B1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance												
	e		P(B wins and 3 points) = 0.7×0.3×0.3 + 0.3×0.7×0.3 or 2 × 0.063 (= 0.126 oe) $\frac{P(\text{B wins \& 3 points})}{P(\text{B wins})} = \frac{0.126}{0.216}$ = $\frac{7}{12}$	M2 M1 A1 [4]	ft their diagram for M-marks soi M1 for one correct product or 0.063 Must attempt $\frac{P(\text{B wins \& 3 points})}{\text{Their (c) NOT (d) or } 0.3 \times 0.3 + 0.7 \times 0.3 \times 0.3 + 0.3 \times 0.7 \times 0.3}$ Allow 0.583 (3 sf) SC no working $\frac{7}{12}$ B2 But minimal working is OK, eg $\frac{2 \times 0.063}{0.216} = \frac{7}{12}$ M2M1A1 Examiner's Comments A common error in the numerator was to use P(A ∩ B) = P(A) × P(B) rather than P(A ∩ B) = P(A) × P(B/A), i.e. in this case the numerator was incorrectly evaluated using 0.216 × 0.42 instead of 2 × 0.3² × 0.7.												
			Total	13													
3	a		<table><tr><td>(55)</td><td>115</td><td>(60)</td><td>(230)</td></tr><tr><td>(65)</td><td>(35)</td><td>70</td><td>170</td></tr><tr><td>120</td><td>(150)</td><td>130</td><td>(400)</td></tr></table>	(55)	115	(60)	(230)	(65)	(35)	70	170	120	(150)	130	(400)	B1 B1 [1]	B1 for 10 correct numbers
(55)	115	(60)	(230)														
(65)	(35)	70	170														
120	(150)	130	(400)														
	b	i	$\frac{55}{400}$ ISW or $\frac{11}{80}$ or 0.1375 or 0.138 (3 sf)	B1 [1]	NB If figures for A in the table are correct, but those for B and C are wrong, correct answers to parts (b)(i), (ii) can be obtained correctly.												
		ii	$\frac{55}{120}$ ISW or $\frac{11}{24}$ or 0.458 (3 sf)	B1 [1]													
	c	i	Kareem because, eg the 2nd digit of each no. is the same as the 1st digit of the next	B1 [1]	oe. Allow Kareem because the numbers are not independent.												
		ii	Only 820 students or 850 > 820	B1 [1]	oe												
			Total	6													

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		DR $\frac{15}{64} \times \frac{2^2}{2!}$ oe eg $(= \frac{15}{32}$ AG) $\frac{15}{64} \times \frac{4}{2}$	B1 (AO1.1) [1]	Must see this expression and result

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
b		DR 2, 2, 5 2, 3, 4 3, 3, 3 $P(X_1 + X_2 + X_3 = 9) =$ $3 \times \left(\frac{15}{32}\right)^2 \times \frac{5}{80} + 6 \times \frac{15}{32} \times \frac{5}{16} \times \frac{5}{32} + \left(\frac{5}{16}\right)^3$ 0.0412 + 0.1373 + 0.0305 $3 \times \frac{225}{16384} + 6 \times \frac{375}{16384} + \frac{125}{4096}$ $\frac{675}{16384} + \frac{1125}{8192} + \frac{125}{4096} \quad (= 0.209045)$ $P(X_1 + X_2 + X_3 = 9 \text{ and at least 1 } X \text{ value} = 2)$ $= 3 \times \left(\frac{15}{32}\right)^2 \times \frac{5}{80} + 6 \times \frac{15}{32} \times \frac{5}{16} \times \frac{5}{32} \quad (= 0.178528)$ $\frac{'0.178528'}{'0.209045'}$ = 0.854 (3 sf) or $\frac{117}{137}$ $P(X_1 + X_2 + X_3 = 9 \text{ and no } X \text{ value} = 2)$ $= \left(\frac{5}{16}\right)^3 \quad (= 0.030518 \text{ or } \frac{125}{4096})$ $1 - \frac{'0.030518'}{'0.209045'}$ = 0.854 (3 sf) or $\frac{117}{137}$	M1 (AO3.1a) M1 (AO2.1) M1 (AO2.1) M1 (AO1.1) M1 (AO2.1) A1 (AO2.2a) M1 M1 A1 [6]	Any two seen, with no more than 2 extra different combinations. eg 0, 4, 5 and 0, 5, 4 count as <u>one</u> extra M2: ≥ 1 correct product actually seen & all three products correct M1: 1 correct product seen or all correct except omission of, or incorrect, multiple(s) or all three results or total correct, but without working Allow M1 for 1 correct product or omit, or incorrect, multiple(s) or ft their probabilities from their previous calculation $\div$ their attempted probs of correct events ft their $P(3, 3, 3)$ $\div$ their attempted probabilities of correct events & subtract from 1 NB $1 - \left(\frac{5}{16}\right)^3$ alone scores M1 <u>Examiner's Comments</u> Most candidates were unable to find their way through the various calculations needed. Some obtained a correct product of three probabilities (e. g. for 2, 2, 5) but omitted to multiply by 3. Others omitted the case 3, 3, 3 altogether.
c		P(two 2's in nine vales of X) or 0.094466 or ${}^9C_2 \times \left(1 - \frac{15}{32}\right)^7 \times \left(\frac{15}{32}\right)^2$ $P(\text{two 2's in nine vales of } X) \times P(X = 2)$	M1 (AO3.1a) M1 (AO2.1) A1 (AO1.1)	soi eg by 9C_2 seen soi

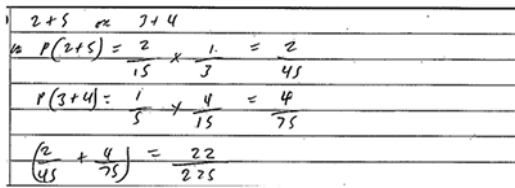
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			or $0.094466 \times \frac{15}{32}$ or ${}^9C_2 \times (1 - \frac{15}{32})^7 \times (\frac{15}{32})^3$ 0.0443 (3 sf)	[3]	NB $(\frac{17}{32})^7 \times (\frac{15}{32})^3$ scores 0, unless multiplied by 9C_2 <u>Examiner's Comments</u> Most candidates did not recognise the need to consider the first 9 values separately before considering the 10th. Some candidates seemed unfamiliar with the binomial distribution.  Misconception In answering Question 7 (about vectors) some candidates appeared not to appreciate the different significances of a, $\mathbf{a}$ and A. Some candidates appeared not to understand the description "cartesian equation" and attempted to find some sort of vector equation of the circle in question 7(d). Many candidates did not appreciate the need for the modulus sign, and how to use it, after integrating $\frac{1}{P-100}$, in question 8. Many candidates used '$=$' instead of '$\leq$' in the hypothesis test, as mentioned in Question 10. In answering Question 13(b), most candidates did not recognise that an infinite series was involved.
			Total	10	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5	a		$k(1 + 2 + 3 + 4 + 5) (= 1)$ $k = \frac{1}{15}$ $P(X = 3) = 3 \times \frac{1}{15} = \frac{3}{15} (= 0.2 \text{ AG})$	M1(AO3.3) A1(AO1.1) A1(AO2.1) [3]	Allow $15k (= 1)$ May be implied Must see $3 \times \frac{1}{15}$ or $\frac{3}{15}$ and answer 0.2	<u>Examiner's Comments</u> Most candidates answered this question correctly. Many, however, (perhaps misunderstanding the phrase "where k is a constant") assumed that all five probabilities were equal, and therefore each probability was 0.2
	b		1 2 3 4 5 $\frac{1}{15}$ $\frac{2}{15}$ $\frac{3}{15}$ $\frac{4}{15}$ $\frac{5}{15}$ oe 0.07, 0.13, 0.2, 0.27, 0.33	M1(AO1.2) A1(AO1.1) [2]	M1 for ≥ 3 probs correct, ft their k cao. Allow decimals (2 dp) SC: Table with all five probs = 0.2: M1	Allow $X = 0$ or $X = 6$ or $X = 6+$ if prob shown as 0 <u>Examiner's Comments</u> Almost all candidates answered this correctly, or incorrectly but consistently with their method for part (a).
	c	i	Both parts. Allow mixture of methods $\frac{3}{15} \times \frac{4}{15} + \frac{2}{15} \times \frac{5}{15}$ oe $\times 2$ $\frac{44}{225}$ or 0.196 (3 sf)	M1(AO3.4) M1(AO3.1a) A1(AO1.1)	Correct products added, ft their table $2 \times (\text{Sum of two products of probs})$ cao Special cases 2-way table Count 4 pairs M1	All probs = 0.2 $0.2^2 + 0.2^2$ M1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
				[3]	But if (b) correct: M0 $\div 2 \ 5$ M1 $= 0.16$ A0	But if (b) correct: M0 $2 \times (0.2^2 + 0.2^2)$ Allow without $2 \times$ M1 $= 0.16$ A0
					Examiner's Comments In both these parts, some candidates either assumed that the probabilities for all the values of X were equal, or (which is equivalent) drew up a two-way table of all the possible outcomes and counted up the relevant ones. These were able to gain only a few marks. Some candidates used one of these incorrect methods, despite having answered parts (a) and (b) correctly. Many candidates recognised that for $T = 7$ could be found by considering $2+5$ and $3+4$, but forgot to consider $5+2$ and $4+3$. Exemplar 1 	
					This candidate has shown clear working, but forgotten to multiply by two in order to consider all the ways to score $T = 7$.	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
		ii	$P(\text{one value is 2 \& } T = 7) = 2 \times \frac{2}{15} \times \frac{5}{15}$ $= \frac{4}{45}$ $\frac{P(\text{one value is 2 \& } T = 7)}{P(T = 7)} \quad \left(\frac{\frac{4}{45}}{\frac{44}{225}} \text{ or } = \frac{0.0889}{0.196} \right)$ $= \frac{5}{11} \text{ or } 0.455 \text{ (3 sf)}$	M1(AO3.4) A1f(AO1.1) M1(AO2.1) A1(AO1.1) [4]	Allow without "2×", ft their table ft their table (except if all probs = 0.2) $\frac{\text{Allow any probability}}{\text{Their (c)(i) or their } P(T = 7)}$ cao NB not 0.454 Eg: If (i) $\frac{22}{225} \div \frac{2}{45} \div \frac{22}{225} = \frac{5}{11}$ M1A0M1A0 Count 2 pairs M1 A0 ÷ 4 M1 = 0.5 A0	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance											
6			Let X be the number of heads. $X \sim B(6, \frac{1}{2})$ $P(D = 0) = P(X = 3)$ $P(D = 2) = 2 \times P(X = 4)$ or similar Values of D are 0, 2, 4, 6 <table border="1"><tr><td>d</td><td>0</td><td>2</td><td>4</td><td>6</td></tr><tr><td>$P(D = d)$</td><td>$\frac{5}{16}$</td><td>$\frac{15}{32}$</td><td>$\frac{3}{16}$</td><td>$\frac{1}{32}$</td></tr></table> The largest value is $\frac{15}{32}$, so the most probable value of D is 2	d	0	2	4	6	$P(D = d)$	$\frac{5}{16}$	$\frac{15}{32}$	$\frac{3}{16}$	$\frac{1}{32}$	M1 (AO3.3) M1 (AO2.1) M1 (AO2.1) B1 (AO1.1) A1 (AO3.4) A1 (AO1.1) A1 (AO1.1) [7]	soi soi soi soi Any two correct All correct For identifying 2	Considering explicitly relationship between D and X
d	0	2	4	6												
$P(D = d)$	$\frac{5}{16}$	$\frac{15}{32}$	$\frac{3}{16}$	$\frac{1}{32}$												
			Total	7												

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																																																											
7		a	(i) <table><tr><td></td><td></td><td colspan="6">Biased die, Y</td></tr><tr><td></td><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td rowspan="6">Fair die, X</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td></tr><tr><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td></tr><tr><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td><td>15</td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr></table> (ii) Four 10s circled or otherwise indicated			Biased die, Y								1	2	3	4	5	6	Fair die, X	1	3	5	7	9	11	13	2	4	6	8	10	12	14	3	5	7	9	11	13	15	4	5	6	7	8	9	10	5	6	7	8	9	10	11	6	7	8	9	10	11	12	B2(AO 1.1a 1.1) [2] B1(AO 1.1) [1]	B1 for ≥ 30 values correct Must be exactly four 10s in table.
		Biased die, Y																																																														
		1	2	3	4	5	6																																																									
Fair die, X	1	3	5	7	9	11	13																																																									
	2	4	6	8	10	12	14																																																									
	3	5	7	9	11	13	15																																																									
	4	5	6	7	8	9	10																																																									
	5	6	7	8	9	10	11																																																									
	6	7	8	9	10	11	12																																																									
		b	Outcomes not equally likely oe	E1(AO 2.3) [1]																																																												
		c	$\frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.1 + \frac{1}{6} \times 0.01$ $= \frac{13}{200}$ or 0.065	M1(AO 3.1b) A1(AO 1.1) [2]																																																												
		d	$P(S = 10 \text{ \& one score} = 4) =$ $\frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.14 + \frac{1}{6} \times 0.01$ $P(\text{One score} = 4 \mid S = 10)$ $= \frac{P(S=10 \text{ \& one score} = 4)}{P(S=10)} = \frac{29}{600} \div \frac{13}{200}$ $= \frac{29}{39}$ or 0.744 (3 sf)	M1(AO 3.1b) M1(AO 2.1) A1(AO 1.1) [3]	ft their (c), dep 1st M1 gained in (d) cao	($= \frac{29}{600}$ or 0.0483(3 sf))																																																										

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
		e	$n = 15$	B1(AO 1.2) [1]		
			Total	10		
8		a	$P(A) \times P(B) = \frac{1}{24}$ $P(A) + P(B) = \frac{1}{24} + \frac{3}{8}$ $P(A) + \frac{1}{24P(A)} = \frac{5}{12}$ $24(P(A))^2 - 10P(A) + 1 = 0$ $(6P(A) - 1)(4P(A) - 1) = 0$ $P(A) = \frac{1}{6}$ and $P(B) = \frac{1}{4}$ or vice versa	M1(AO1.1a) M1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1) [5]	Attempt equation in one P Correct quadratic equation in one P Allow without "vice versa"	
		b	$P(C') = 1 - P(C)$ (= 0.4) $P(D \cap C') = P(C \cup D) - P(C)$ (= 0.2) $P(D C') = \frac{P(D \cap C')}{P(C')}$ $= \frac{0.2}{0.4} = 0.5$	M1(AO 1.2) M1(AO 1.1) M1(AO 1.1) M1(AO 1.2) [4]	Attempted	
			Total	9		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
9		a	Attempt to represent information e.g. by Venn diagram with x in centre and 3 other correct values in terms of x Attempt total (in terms of x) = 30 $x = 4$ so $n(S \cap H \cap T) = 4$	B1(AO3.3) M1(AO3.4) E1(AO1.1) [3]	Any equivalent method Or the number doing all three is 4. E0 for just $x = 4$	OR B1 $\frac{18}{30} + \frac{19}{30} + \frac{17}{30} - \left(\frac{8}{30} + \frac{9}{30} + \frac{11}{30} \right) = \frac{26}{30}$ M1 $1 - \frac{26}{30} = \frac{4}{30}$
		b	 oe	B1FT(AO2.2a) [1]	FT their (a)	
		c	$\frac{5}{9} \times \frac{19}{29}$ $\frac{4}{9} \times \frac{18}{29}$ $\frac{5}{9} \times \frac{19}{29} + \frac{4}{9} \times \frac{18}{29}$ $= \frac{167}{261} \text{ oe or } 0.640 \text{ (3 s.f.)}$	B1(AO2.2a) B1(AO2.2a) M1(AO2.2a) A1(AO1.1) [4]	All correct	
Total				8		

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Question			Answer/Indicative content	Marks	Guidance	
10		a	$P(X > 39) = P(X = 40) = \frac{1}{860}(1 + 40)$ $= \frac{41}{860}$	M1(AO1.1) A1(AO1.1) [2]	Attempt at evaluating $P(X = 40)$	
		b	$P(X \text{ even}) = \frac{1}{860}(20 + (2 + 4 + 6 + \dots + 40)) \text{ oe}$ $= \frac{1}{860}\left(20 + \frac{2+40}{2} \times 20\right)$ $= \frac{22}{43}$ $P(X = 2, 4, 6, 8) = \frac{1}{860}(4 + 2 + 4 + 6 + 8)$ $= \frac{12}{430} \text{ oe}$ $\frac{P(X = 2, 4, 6, 8 \text{ and } X \text{ even})}{P(X \text{ even})} = \frac{P(X = 2, 4, 6, 8)}{P(X \text{ even})}$ $= \frac{12}{430} \div \frac{22}{43} = \frac{3}{55} \text{ oe or } 0.0545 \text{ (3 s.f.)}$	M1(AO3.1a) A1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO3.2a) B1(AO2.1) [6]	Attempt Σ probabilities of all even values Correct expression Attempt Σ probabilities for $X = 2, 4, 6, 8$ their $P(X = 2, 4, 6, 8)$ their $P(X \text{ even})$ For a clear solution allowing the line of reasoning to be followed, with each component of the conditional probability found clearly	Numerical sums may be evaluated BC throughout
			Total	8		